# Sentinel v1

## Dashboard Build Brief

*Prepared for Anshul / Frontly · Booked Digital Ltd*

## 1. Overview

Sentinel is a single pane of glass for the principal of a UK dental practice. Every view answers one question: "what is happening in my practice and what needs my attention?" The product replaces three to four tools the practice currently jumps between (PMS reports, call provider dashboard, marketing platform, ad-hoc spreadsheets) with one editorial-style dashboard.

v1 ships four views: Calls, Patients, Patient Journey, and Revenue. v2 adds the Ask Sentinel natural-language layer on top of the same data — out of scope here but designed for in v1.

### Design philosophy

- Information density without clutter. Closer to Linear or a Bloomberg terminal than a typical SaaS dashboard.
- Dark editorial aesthetic, amber accents, generous whitespace, restrained typography.
- Every number is clickable and drills down to the underlying records.
- Empty states matter. If a data source isn't connected, say so clearly with a setup CTA — never show zeros that look like a bug.
- The principal should land on any view and have the answer they need within five seconds.

### Global controls

- Date picker (top right of every view): Today, 7d, 30d, 90d, Custom. Default 30d. Drives every widget on the page.
- Practice selector (visible only for users with access to multiple practices).
- CSV export available on every table and chart.
- View state persists in URL params so views are shareable and bookmarkable.

### Architecture notes

- Multi-tenant from day one. Every query scoped by practice_id
- All metrics computed by deterministic functions. v2's Claude analyst layer calls these functions via tool-use and never computes numbers directly.
- Auth . Roles: principal (full access), practice manager (full access), front-desk (calls + patients only).
- Data sources: Dentally (PMS), GoHighLevel (CRM + human call data), frontly (AI calls), Twilio (call telephony where applicable).

## 2. Calls Dashboard

Purpose: a unified view of every call into the practice — AI and human — with a toggle to slice the data three ways. The principal should be able to see at a glance whether calls are being handled, by whom, and what they are producing.

### Source toggle

Top of page, segmented control: All | AI | Human. Default to All. The toggle drives every widget on the page — KPIs, charts, table. State persists in URL params.

### KPI strip

Six KPIs across the top:

- Total calls
- Answered rate
- Missed calls
- Appointments booked
- Revenue booked (sum of treatment plan values from booked appointments)
- Average handle time

When All is selected, each KPI shows a small AI-vs-human split underneath (e.g. "127 calls — 84 AI / 43 human"). When toggled to AI or Human, the split disappears and numbers reflect only that source.

### Side-by-side comparison block (only visible when toggle = All)

Two columns, AI on the left, Human on the right, same metrics in the same order: calls handled, booking rate, revenue booked, missed/abandoned, average duration. The visual job is to make it instantly obvious where each channel is winning and losing. This is the most important block on the page when both sources are in view.

### Charts

- **Time-series of calls per day.** Three series: AI, human, total. Stacked area when toggle = All, single line when filtered. Legend allows toggling individual series.
- **Hour-of-day heatmap (calls by hour × day-of-week).** Shows when calls are coming in and when they are being handled or missed. When toggle = All, two heatmaps side by side, AI and human, same colour scale so they are directly comparable.
- **Outcome breakdown — donut or horizontal bar.** Split by booked, info-only, escalated, missed/abandoned. When toggle = All, two donuts side by side.

### Unified call log (table)

Single table at the bottom, every call regardless of source. Columns:

- Timestamp
- Source (AI / Human badge — colour-coded so the table is scannable at a glance)

- Caller number
- Caller name (if matched to a Dentally patient record)
- Duration
- Intent detected
- Outcome
- Appointment created (link to Dentally appointment if one was booked)
- Recording (AI only, inline player on click)
- Transcript (AI only, expanded on click)

Filterable by source, outcome, practitioner requested, treatment type, date range.

## 3. Patients

Purpose: a working view of the patient base, not a static list. This is the spine for every reactivation and follow-up workflow.

**Headline metrics**

- Total active patients
- New patients this period
- Lapsed patients (no appointment in 12+ months)
- Reactivation candidates (lapsed but with prior treatment value above a configurable threshold — default £500)

**Breakdowns**

- NHS vs private (count and revenue)
- By primary practitioner
- By membership plan status (hooks into Veroplan in a later release — show as "Plan" / "No plan" in v1)

**Patient table**

Searchable, sortable, filterable. Columns:

- Name
- Date of birth / age
- Last visit
- Next appointment (if booked)
- Lifetime value
- Outstanding treatment plan value
- Membership status
- Last contact channel + date

Click into a patient → opens the patient timeline (see Patient Journey, individual timeline section). The Patient Journey view and the Patients view share this drill-down — same component, two entry points.

## 4. Patient Journey

Purpose: show the full lifecycle of a patient from first touch through every subsequent treatment, so the principal can see which acquisition channels produce patients who actually return and spend more over time. This is the view that turns "cost per lead" into "lifetime value by source."

This is the differentiating view in v1 — no UK dental software currently exposes this. It is also the foundation for the Ask Sentinel chat layer in v2 ("which marketing channel produced the highest 12-month LTV last year?" becomes a one-line answer).

#### Cohort overview (top of page)

Cohort table grouped by acquisition month and source. Rows are cohorts (e.g. "March 2026 — Meta Ads — Invisalign campaign"). Columns are months 0, 1, 3, 6, 12, 24.

Cell values default to cumulative revenue per patient in that cohort by that month. Toggle at top of widget switches between:

- Cumulative revenue per patient
- Average revenue per patient
- Retention rate (% of cohort with any visit in that month)

Colour intensity scales with value — heatmap-style — so the principal can immediately see which cohorts are compounding and which are flat after the first treatment.

Filters: acquisition source (Meta, Google, AI call, walk-in, referral, reactivation), initial treatment type, practitioner.

#### Treatment flow (Sankey diagram)

Visual flow showing how patients move through treatment categories over time.

- Left-most column: acquisition source
- Next column: first treatment
- Subsequent columns: second treatment, third treatment, etc.
- Width of each flow band represents patient count
- Hover shows revenue, average days between treatments, and patient count
- Click any band to see the patient list behind it

This is the view that answers "of patients who came in for Invisalign, how many came back for whitening, and how many months later on average?"

#### Individual patient timeline

Click any patient — from the cohort table, the Sankey, or the Patients view — and get a vertical timeline of their entire history with the practice. Every entry is a node on the timeline:

- Acquisition event (lead source, first call/form fill, marketing campaign attribution)
- First contact (AI or human call, with link to recording / transcript)
- First appointment booked
- Treatment plan created (with value)
- Each appointment attended
- Each payment
- Each subsequent treatment plan
- Reactivation touches (campaign sends, AI follow-up calls)
- Lapses (gaps > 12 months flagged in red)

Each node shows date, type, value if applicable, and clicks through to the source record (call recording, Dentally appointment, GHL contact, etc.). The timeline visually compresses or expands based on activity density — busy months stretch out, quiet years collapse. Cumulative lifetime value runs along the side as a running line.

**Headline metrics (above the cohort table)**

- Average lifetime value by acquisition source
- Average treatments per patient by source
- Average months between first and second treatment
- % of patients who return for a second treatment within 12 months
- Best-performing source by 12-month LTV

## 5. Revenue — New vs Existing Treatment Plans

Purpose: separate "growth" revenue from "in-flight" revenue so the principal can see whether the top of the funnel is actually feeding through. Most practices look at one revenue number and miss the leading vs lagging signal.

**Primary chart**

Single chart, monthly bars, side-by-side or stacked (toggle):

- Bar A: revenue from newly created treatment plans in that month
- Bar B: revenue from previously existing plans completed in that month

Default view: 12-month rolling. Toggle for 6m / 12m / 24m.

**Supporting metrics**

- % of monthly revenue from new vs existing
- Average treatment plan value (new)
- Average days from plan creation to first payment

- Total new plan value created (leading indicator)
- Total existing plan value cleared (lagging indicator)

**Drill-down**

Click any bar → table of the underlying treatment plans. Columns: patient, practitioner, treatment, value, date created, date completed, status. Each row links through to the patient timeline.

## 6. Cross-Cutting Requirements

**Interaction**

- Every number on every page is clickable and drills to the underlying records.
- Every chart respects the global date picker.
- Every table has CSV export.
- Every view has a clear empty state with a setup CTA where data isn't connected.

**Performance**

- Initial dashboard load under 2 seconds for a practice with 12 months of data.
- All aggregations precomputed where possible (nightly refresh) and queried via the deterministic Supabase functions; only fine-grained drill-downs hit live data.

**Visual language**

- Dark theme by default. Background near-black, surfaces in dark grey, text in soft white.
- Amber as the single accent colour for emphasis, alerts, and key numbers.
- Numerical typography: tabular figures, slightly oversized for hero KPIs.
- Charts: minimal grid, subtle gridlines, labels only where they earn their place.
- No drop shadows, no gradients, no rounded illustrations. Editorial restraint throughout.

**Responsive**

- Primary target: desktop, 1440px and above.
- Tablet (1024px): graceful degradation, side-by-side blocks stack.
- Mobile: read-only summary view of KPIs and headline charts only — full dashboard is not a mobile use case.

**Auth and access**

- Supabase Auth, magic link login as the default.
- Row-level security on every table, scoped by practice_id.
- Roles: principal, practice manager, front-desk.

## 7. Out of Scope for v1

- Pipeline / treatment plan kanban (deferred — possibly returns in v1.1)

- Ask Sentinel natural language chat layer (v2)
- Veroplan membership plan deep integration (v1 surfaces only the Plan / No plan flag in Patients)
- Compliance module (separate product, separate roadmap)
- Marketing campaign send/composition (v1 reads campaign attribution only, does not author campaigns)

## 8. Suggested Build Sequence

Recommended order to get a usable beta on scre:

- **1. Auth, multi-tenancy, RLS, base layout, navigation, global date picker.**
- **2. Calls Dashboard** (highest immediate value — AI vs human comparison is the wedge).
- **3. Patients view** (table + drill-down — feeds the timeline component).
- **4. Revenue view** (new vs existing plans — single chart, fast to ship).
- **5. Patient Journey** (cohorts → Sankey → individual timeline — the most ambitious view, ship last).

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